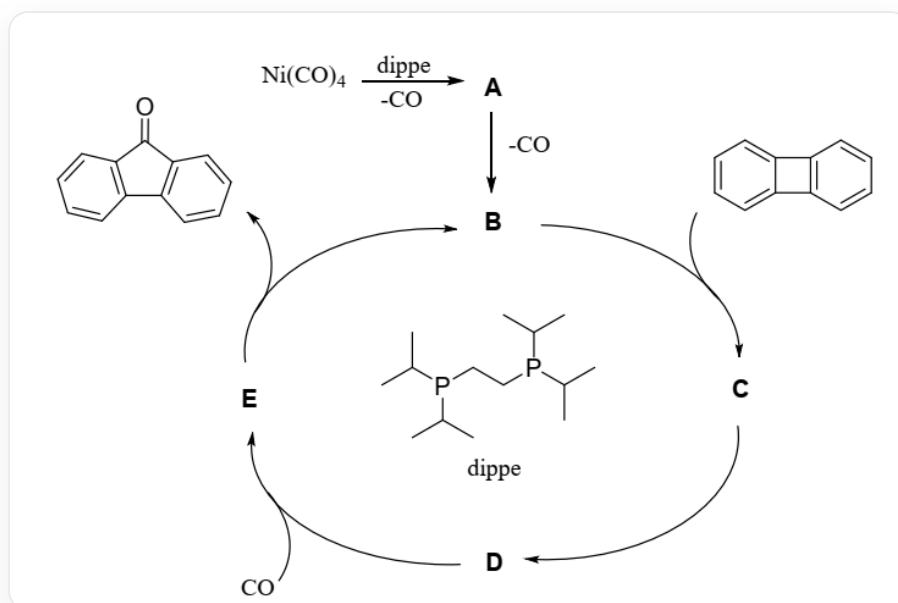


## Question

CO, as a common waste product in industrial production, has been a focal point of research for the preparation of carbonyl compounds. A laboratory achieved the reuse of CO using dippe and  $\text{Ni}(\text{CO})_4$  as catalysts.  $\text{Ni}(\text{CO})_4$  and dippe undergo a ligand substitution reaction to form the precursor **A**. Under a CO atmosphere, biphenylene is added, and the reaction catalyzed by  $\text{Ni}(\text{CO})_4$  and dippe yields dibenzocyclopentenone. The catalytic cycle is illustrated in the following diagram:



The diagram depicts a reaction system with a cyclic structure, involving 6 reactions. The first reaction is  $\text{O}[\text{C}][\text{Ni}][\text{C}][\text{O}][\text{C}][\text{O}][\text{C}][\text{O}] > \text{dippe} > [\text{A}]$ , releasing one molecule of  $\text{C}\#\text{O}$ ; the second reaction is  $[\text{A}] >> [\text{B}]$ , releasing another molecule of  $\text{C}\#\text{O}$ . The subsequent four reactions form a cyclic structure in the diagram:

$[\text{B}] > \text{C1}=\text{CC}=\text{C2C}(=\text{C1})\text{C3}=\text{C2C}=\text{CC}=\text{C3} > [\text{C}]$ ,  $[\text{C}] >> [\text{D}]$ ,  $[\text{D}] > \text{C}\#\text{O} > [\text{E}]$ ,  $[\text{E}] >> [\text{B}]$ .  $\text{C1}=\text{CC2}=\text{C}(\text{C}=\text{C1})\text{C}(=\text{O})\text{C3}=\text{C2C}=\text{CC}=\text{C3}$ . Here, all instances of "[C]" represent the substance code C rather than methane. The structure of dippe is CC(C)P(CCP(C(C)C)C(C)C)C(C)C

Identify all steps involving the formation of new carbon-carbon bonds in this reaction cycle.

**A.** Only from **B** to **C**.

**B.** Only from **C** to **D**.

**C.** Only from **D** to **E**.

**D.** Only from **E** to **B**.

**E.** From **B** to **C**, and from **C** to **D**.

**F.** From **B** to **C**, and from **D** to **E**.

**G.** From **B** to **C**, and from **E** to **B**.

**H.** From **C** to **D**, and from **D** to **E**.

**I.** From **C** to **D**, and from **E** to **B**.

**J.** From **D** to **E**, and from **E** to **B**.

**K.** From **B** to **C**, from **C** to **D**, and from **D** to **E**.

**L.** From **B** to **C**, from **C** to **D**, and from **E** to **B**.

**M.** From **B** to **C**, from **D** to **E**, and from **E** to **B**.

**N.** From **C** to **D**, from **D** to **E**, and from **E** to **B**.

**O.** From **B** to **C**, from **C** to **D**, from **D** to **E**, and from **E** to **B**.

**P.** There are no steps involving the formation of new carbon-carbon bonds in this reaction cycle.

## Answer

Correct Answer: **I**

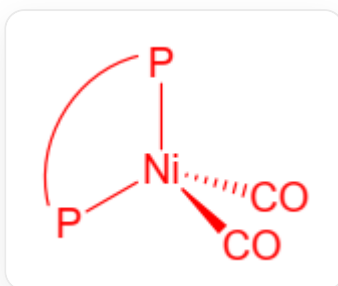
## Detailed Explanation

From  $\text{Ni}(\text{CO})_4$  to **A**, a ligand substitution reaction occurs, where the bidentate ligand dippe replaces two CO ligands, resulting in the tetrahedral complex **A** with the structure  $(\text{dippe})\text{Ni}(\text{CO})_2$ .

### CHECKPOINT

1 PTS

The transformation from  $\text{Ni}(\text{CO})_4$  to **A** is a ligand substitution reaction



CC(C)[P]1(C(C)C)[Ni]([P](C(C)C)(C(C)C)CC1)([C]#O)[C]#O, i.e., the structure of  $(\text{dippe})\text{Ni}(\text{CO})_2$ , tetrahedral

In the **A** molecule, Ni is 4-coordinate.

### CHECKPOINT

1 PTS

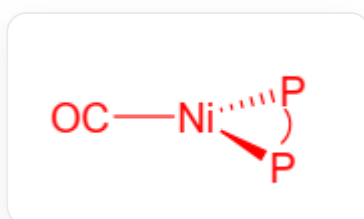
**A** is  $(\text{dippe})\text{Ni}(\text{CO})_2$

From **A** to **B**, one carbonyl ligand dissociates, changing the geometry to trigonal planar, with **B** being (dippe)Ni(CO).

## CHECKPOINT

1 PTS

The transformation from **A** to **B** is a ligand dissociation reaction



CC(C)[P]1(C(C)C)[Ni]([P](C(C)C)(C(C)C)CC1)[C]#O, i.e., the structure of (dippe)Ni(CO), trigonal planar

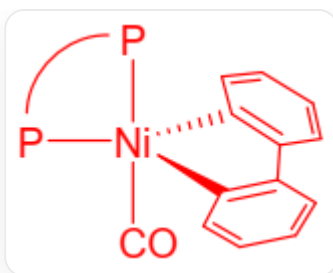
In the **B** molecule, Ni is 3-coordinate.

## CHECKPOINT

1 PTS

**B** is (dippe)Ni(CO)

**B** undergoes an oxidative addition reaction with the organic molecule C1=CC=C2C(=C1)C3=C2C=CC=C3 to yield **C**. In this process, one C – C bond is broken, and two new C – Ni bonds are formed, resulting in a five-membered ring. The structure of **C** is:



CC(C)[P]1(C(C)C)[Ni]2([P](C(C)C)(C(C)C)CC1)(C3=CC=CC=C3C4=CC=CC=C42)[C]#O, where the complex center consists of one Ni atom, coordinated by one monodentate CO, one bidentate dippe, and one bidentate C1=CC=C[C]=C1C2=[C]C=CC=C2. The geometry is trigonal bipyramidal, with the carbon of C#O and one phosphorus atom of dippe in axial positions, while the remaining coordination atoms lie in the equatorial plane.

### CHECKPOINT

1 PTS

The transformation from **B** to **C** is an oxidative addition reaction

In the **C** molecule, Ni is 5-coordinate with a trigonal bipyramidal geometry.

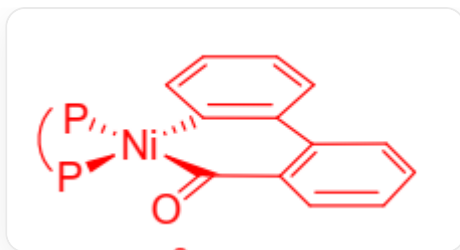
From **C** to **D**, a migratory insertion reaction occurs intramolecularly, breaking one C – Ni bond and forming one C – C bond.

### CHECKPOINT

1 PTS

The transformation from **C** to **D** is a migratory insertion reaction

The structure of **D** is:



CC(C)[P]1(C(C)C)[Ni]2(C3=CC=CC=C3C4=CC=CC=C4C2=O)[P](C(C)C)(C(C)C)CC1, where the complex center consists of one Ni atom, coordinated by one bidentate dippe and one bidentate [C](=O)C1=C(C=CC=C1)C2=[C]C=CC=C2. The geometry is square planar.

In the **D** molecule, Ni is 4-coordinate.

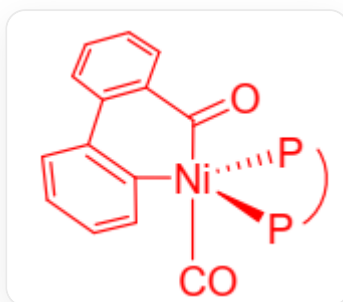
From **C** to **D**, one new carbon-carbon bond is formed.

### CHECKPOINT

1 PTS

From **C** to **D**, one new C-C bond is formed

From **D** to **E**, one molecule of CO is added, forming one new C – Ni bond while the structures of the other ligands remain unchanged. The structure of **E** is:



CC(C)[P]1(C(C)C)[Ni]2(C3=CC=CC=C3C4=CC=CC=C4C2=O)([C]#O)[P](C(C)C)(C(C)C)CC1, where the complex center consists of one Ni atom, coordinated by one monodentate C#O, one bidentate dippe, and one bidentate [C](=O)C1=C(C=CC=C1)C2=[C]C=CC=C2. The geometry is trigonal bipyramidal, with the carbon of C#O and the acyl carbon of [C](=O)C1=C(C=CC=C1)C2=[C]C=CC=C2 in axial positions, while the remaining coordination atoms lie in the equatorial plane.

In the **E** molecule, Ni is 5-coordinate.

### CHECKPOINT

1 PTS

The transformation from **D** to **E** is a ligand association reaction

From **E** to **B**, a reductive elimination reaction occurs, releasing one molecule of C1=CC2=C(C=C1)C(=O)C3=C2C=CC=C3, and Ni reverts from 5-coordinate to 3-coordinate.

### CHECKPOINT

1 PTS

The transformation from **E** to **B** is a reductive elimination reaction

In this process, a new carbon-carbon bond is formed in the released organic molecule C1=CC2=C(C=C1)C(=O)C3=C2C=CC=C3.

### CHECKPOINT

1 PTS

From **E** to **B**, a new C-C bond is formed in the released organic molecule

In summary, during the catalytic cycle described above, new carbon-carbon bond formation occurs in the transformations from **C** to **D** and from **E** to **B**.